TASK 2: PREDICTIVE MODELING USING MACHINE LEARNING

Internship Project Report – Thiranex EduTech

1. Introduction

This project demonstrates predictive modeling using machine learning classification algorithms. The objective is to classify employee salary levels into High and Low categories using employee-related features.

2. Objectives

• Prepare and preprocess an employee dataset.

• Train multiple machine learning classification models.

• Evaluate models using accuracy, confusion matrices and ROC-AUC.

• Compare model performance and identify the strongest result.

3. Dataset Description

A synthetic dataset containing 300 employee records was generated for this demonstration. It contains Age, Salary, Department, Gender, Join_Date and Salary_Level. Salary_Level was created using the median salary as the classification threshold, producing 150 High and 150 Low records.

Important: The dataset is synthetic and does not represent real employees. The evaluation results demonstrate the workflow only and should not be interpreted as real-world predictive performance.

4. Methodology

• Created a synthetic dataset of 300 records.

• Derived Experience from Join_Date.

• Converted categorical features into numerical dummy variables.

• Separated features and target variable.

• Split the dataset into 80% training and 20% testing data.

• Applied feature scaling for Logistic Regression.

• Trained Logistic Regression, Decision Tree and Random Forest.

5. Algorithms Used

Logistic Regression: A classification algorithm that estimates the probability of belonging to a class.

Decision Tree: A tree-based algorithm that makes predictions using feature-based decision rules.

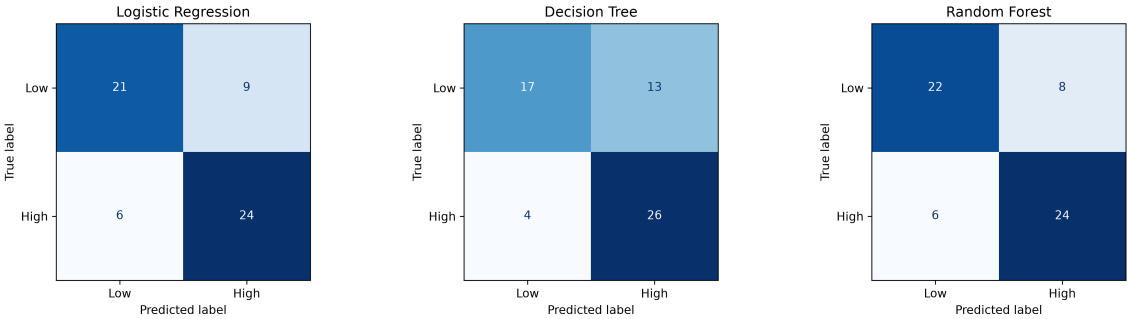
Random Forest: An ensemble algorithm that combines multiple decision trees to produce predictions.

6. Model Performance Comparison

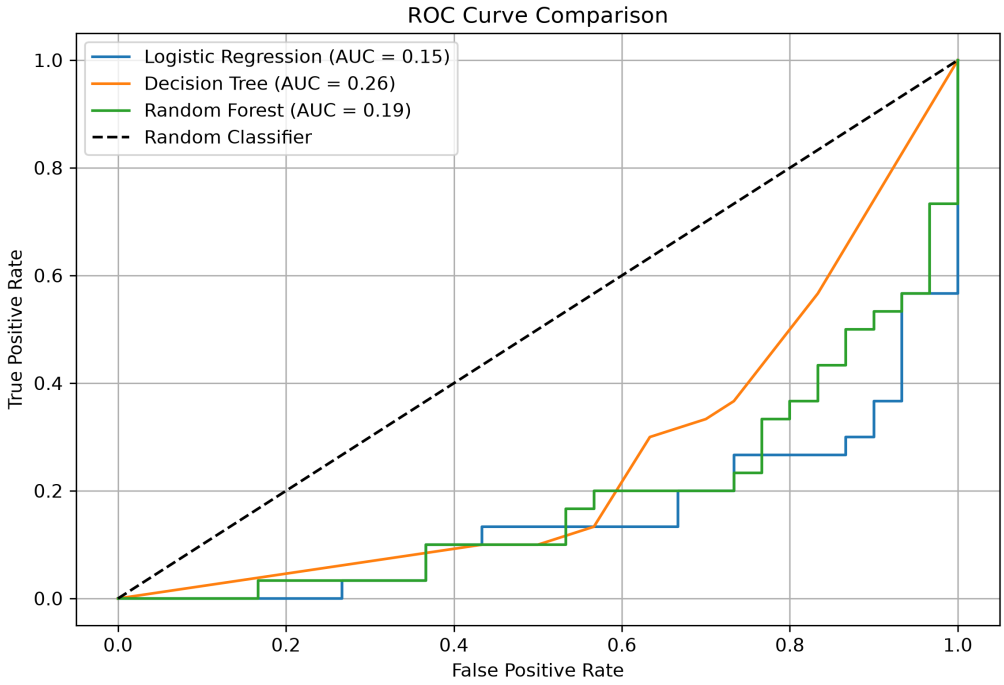
Model	Accuracy (%)	AUC Score
Logistic Regression	75.00	0.8500
Decision Tree	71.67	0.7394
Random Forest	76.67	0.8144

Random Forest achieved the highest accuracy of 76.67%, while Logistic Regression achieved the highest AUC score of 0.8500.

7. Confusion Matrices



8. ROC Curve Comparison



9. Classification Reports

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==================================================
Logistic Regression
==================================================
precision recall f1-score support

High 0.73 0.80 0.76 30
Low 0.78 0.70 0.74 30

accuracy 0.75 60
macro avg 0.75 0.75 0.75 60
weighted avg 0.75 0.75 0.75 60


==================================================
Decision Tree
==================================================
precision recall f1-score support

High 0.67 0.87 0.75 30
Low 0.81 0.57 0.67 30

accuracy 0.72 60
macro avg 0.74 0.72 0.71 60
weighted avg 0.74 0.72 0.71 60


==================================================
Random Forest
==================================================
precision recall f1-score support

High 0.75 0.80 0.77 30
Low 0.79 0.73 0.76 30

accuracy 0.77 60
macro avg 0.77 0.77 0.77 60
weighted avg 0.77 0.77 0.77 60
```

10. Conclusion

Three classification algorithms were successfully trained and evaluated. Random Forest achieved the highest test accuracy (76.67%) and F1-score (approximately 0.77), while Logistic Regression achieved the highest AUC score (0.8500). These results are limited to the synthetic dataset and its generated patterns. Further evaluation on a larger, representative real-world dataset would be required before practical use.